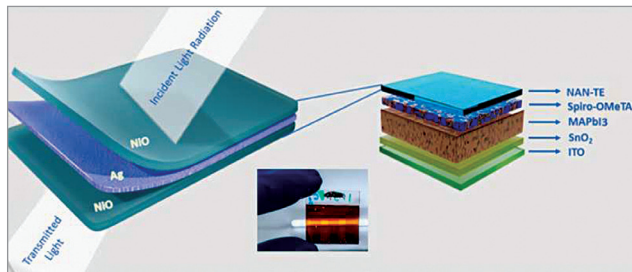




Durable Bifacial Perovskite Cells

Researchers at IIT Dharwad have advanced bifacial perovskite solar cell technology by developing a transparent electrode that enhances efficiency, durability, and infra-red transparency. Their hybrid three-layer NiO/Ag/NiO



High-efficiency, durable bifacial perovskite solar cells (Credit: www.spie.org)

(NAN) electrode, created using low-energy physical vapour deposition, offers minimal resistance and high visible light transmission. The solar cells achieved 9.05% and 6.54% power conversion efficiency from front and rear illumination, with a 72% bifaciality factor. Retaining 80% efficiency after 1000+ hours, these ultra-thin ($\leq 40\text{nm}$) electrodes enable integration into building materials, agrivoltaics, and advanced optoelectronic applications.

Hybrid Microscope For 3D Imaging

Marine Biological Laboratory researchers have developed a hybrid microscope combining polarised fluorescence technology with a dual-view light sheet microscope (diSPIM). This innovation allows, for the first time, complete 3D orientation and position mapping of molecules like labeled proteins inside cells. Traditional polarised light microscopy struggles with tilted spindles, leading to ambiguous read-outs. The new microscope corrects for tilt, enabling precise imaging of microtubules.

Bridge Interlayer For 2D Optoelectronics

Scientists in South Korea have developed a conductive-bridge interlayer contact, overcoming efficiency limits in 2D semiconductor-based photodiodes. These devices, crucial for imaging, sensing, and communication, suffer from Fermi level pinning, which disrupts charge flow. A team from KIST, Korea University and Yonsei University designed an insulating oxide interlayer with gold nanoclusters, enabling Fermi level depinning and low resistance. Applied to a WS_2 photodiode, it achieved 0.29A/W photoresponsivity, 122dB linear dynamic range, and 9.9% power efficiency, advancing 2D optoelectronics for imaging and sensing.

Shape-Shifting Robot For Any Terrain

GOAT (Good Over All Terrains), developed by Josie Hughes' team at EPFL's CREATE Lab, is a shape-shifting robot designed for versatile movement. Inspired by animals like spiders and octopuses, GOAT transitions between a flat 'rover'

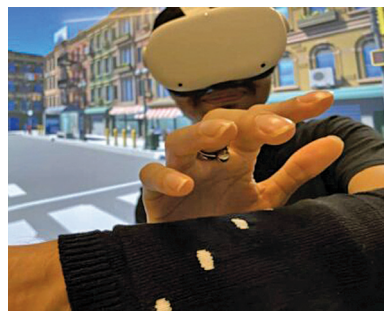


The morphing GOAT robot in sphere mode (Credit: CREATE EPFL)

mode and a spherical shape, enabling it to roll, drive, and swim with minimal energy use. Made from fibreglass rods and motorised wheels, its frame morphs via winch-driven cables. GOAT relies on GPS and an inertial measurement unit, operating without cameras. Future research aims to refine algorithms and explore disaster response and space exploration applications.

Washable Smart Textiles With Magnetic Sensors

Researchers from Nottingham Trent University, Helmholtz-Zentrum Dresden-Rossendorf, and Free University of



The textile can be operated by the user through the use of a ring or glove (Credit: Free University of Bozen-Bolzano)

Bozen-Bolzano have developed washable, durable e-textiles that detect magnetic fields. These textiles enable touchless interaction via a magnet in a ring or glove. Embedded sensors, marked with dye or embroidery, function as buttons or controls.

The technology supports textile-based keyboards, resists accidental activation, and operates underwater. Applications include VR navigation, smart clothing, and interactive fashion. Unlike capacitive sensors, it remains reliable in diverse conditions, offering a breakthrough in wearable human-computer interaction.